LESSON 6.14

The Distributive Property

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the distributive property to expand and factor expressions.

WHY THIS MATTERS You're selling concert bundles at the music studio. Each bundle includes a ticket ($15) and a snack ($5). If 3 people each buy a bundle, you can calculate the total as 3(15 + 5) OR as 3 × 15 + 3 × 5. Either way, the total is the same — $60! This is the distributive property.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
The Distributive Property *Spanish: La propiedad distributiva*	Multiply a factor by every term inside parentheses.	3(x + 4) = 3x + 12
Distributive Property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	3(4 + 5) = 3×4 + 3×5 = 12 + 15 = 27 — the 3 gets 'distributed' to both the 4 and the 5
Factor *Spanish: Factor*	A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because 12 ÷ 3 = 4.	In 3(x + 2), the 3 is the factor outside the parentheses that multiplies each term inside
Expand *Spanish: Desarrollar*	To multiply out the parentheses in an expression.	Expand 2(n + 6): multiply 2 × n = 2n and 2 × 6 = 12, so 2(n + 6) = 2n + 12
Equivalent *Spanish: Equivalente*	Expressions that always have the same value.	3(x + 4) and 3x + 12 always give the same answer: when x = 2, both = 18; when x = 10, both = 42
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 6x + 15, the coefficient of x is 6 — it came from distributing in 3(2x + 5)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Each bundle has a ticket ($15) and a snack ($5). For 3 bundles I can write 3(15 + 5).

1. Way 1: add inside first. 15 + 5 = 20, then 3 × 20 = 60.
2. Way 2: distribute the 3 to each part. 3 × 15 + 3 × 5.
3. That is 45 + 15 = 60.
4. Both ways give $60, so 3(15 + 5) = 3 × 15 + 3 × 5.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's expand 2(8 + 3). Where does the 2 go?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I distribute, I multiply the outside factor _____ by each term _____. The operation inside the parentheses _____ stays the same.

Word bank The Distributive Property | Distributive Property | Factor | Expand | Equivalent | Coefficient | distribute | multiply | each term | factor | expand | equivalent

WATCH OUT!

A common mistake

The most common mistake in The Distributive Property is multiplying the outside factor by only the FIRST term inside the parentheses and forgetting the second

LESSON 6.14

The Distributive Property

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Expand 6(n + 3) using the distributive property.

(A) 6n + 18
(B) 6n + 3
(C) 6n + 9
(D) n + 18

STEPS TO TRY
1. The 6 outside must visit BOTH terms inside the parentheses: the n and the 3.
2. Multiply 6 × n and 6 × 3 separately, then join the two products with the plus sign.
3. First product: 6 × n = 6n. Now find the second product 6 × 3 and add it on.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Expand 4(5 − 2) using the distributive property.

(A) 9

(B) 12

(C) 20 − 2

(D) 22

STEPS TO TRY

1. Even with subtraction inside, the 4 must multiply BOTH the 5 and the 2.
2. Find 4 × 5 and 4 × 2 separately, then subtract the second product from the first.
3. First product: 4 × 5 = 20. Next multiply 4 × 2, then subtract that product from 20.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Expand 3(x + 4) using the distributive property.

(A) 3x + 12

(B) 3x + 4

(C) 7x

(D) x + 12

STEPS TO TRY

1. The 3 outside multiplies each term inside — both the x and the 4.
2. Write two products: 3 times x, and 3 times 4. Connect them with the + from inside.
3. Begin: 3 × x = 3x. Now multiply 3 × 4 and attach that product with a plus sign.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

When using the distributive property with a(b + c), you must multiply 'a' by:

(A) Only b

(B) Only c

(C) b + c first, then multiply

(D) Both b and c

STEPS TO TRY

1. This asks about the RULE itself: where does the outside factor 'a' go?
2. Picture arrows: 'a' reaches over the parentheses to every term it touches. How many arrows are there?
3. Test with numbers: 2(3 + 4) = 14. Try multiplying 2 by only the 3, then adding 4 — do you still get 14?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

SENTENCE STARTER When I distribute, I multiply the outside factor ___ by each term ___. The operation inside the parentheses ___ stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help